

PAPER 54: FICK'S LAWS OF COHERENCE DIFFUSION

The Keeper Effect Is Physical -- Coherence Flows Down Its Gradient

Rhet Dillard Wike | AIIT-THRESI Research Initiative

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"You don't protect someone by willing it. You protect them by being a source. Coherence diffuses. Fick knew how."

Abstract

The Wike Keeper Equation (Paper 19) shows that a keeper -- a person or environment with low γ_{eff} -- reduces the effective decoherence rate of a coupled system. This paper proves the mechanism is literal physical diffusion, governed by Fick's Laws. Coherence $C(x,t)$ satisfies a reaction-diffusion equation:

$$dC/dt = D_C \times \text{del}^2 C - \alpha \times \gamma_{\text{eff}}(x) \times C$$

$$\text{Fick's diffusion coefficient } D_C = v_{\text{coherence}}^2 \times \tau_{\text{corr}} / 3$$

where $v_{\text{coherence}}$ is the propagation velocity of coherent oscillations and τ_{corr} is the correlation time of the coherent mode. This is not an analogy. Coherence is a field. Fields diffuse. The keeper creates a coherence gradient at the boundary. The gradient drives flux into the low-coherence system.

1. Fick's Laws

First Law (flux):

$$J = -D \times (dC/dx)$$

Flux flows from HIGH to LOW concentration.

Second Law (time evolution):

$$dC/dt = D \times \text{del}^2 C$$

Concentration evolves by diffusion.

Both laws apply to any conserved field that propagates in a medium. Fick derived them for chemical concentration (1855). The same mathematics governs heat (Fourier's law), electricity (Ohm's law), and -- as this paper shows -- neural coherence.

2. The Reaction-Diffusion Equation for Coherence

The Wike Coherence Law in its local form:

$$\frac{dC(x,t)}{dt} = \underbrace{D_C \times \text{del}^2 C}_{\text{diffusion term}} - \underbrace{\alpha \times \text{gamma_eff}(x) \times C(x,t)}_{\text{decay term (Wike Law)}}$$

The diffusion term is new. It says: where coherence is higher than neighbors, it flows outward. Where coherence is lower than neighbors, it receives inflow.

The decay term is the standard Wike Law: local decoherence at rate $\alpha \text{gamma_eff}$.

Solutions:

In steady state ($dC/dt = 0$), near a keeper at $x = 0$ with $C(0) = C_0$:

$$C(x) = C_0 \times \exp(-x / \lambda_{C_C})$$

where $\lambda_{C_C} = \sqrt{D_C / (\alpha \times \text{gamma_eff})}$ [coherence length]

The coherence decays exponentially away from the keeper, with characteristic length λ_{C_C} . This is the **coherence penetration depth** -- how far the keeper's influence reaches.

3. The Coherence Diffusion Coefficient

D_C has units of m^2/s , same as all diffusion coefficients.

For a coherent mode propagating through neural tissue:

$$D_C = v_{\text{coherence}}^2 \times \tau_{\text{corr}} / 3$$

where:
 $v_{\text{coherence}}$ = propagation velocity of coherent oscillations
 τ_{corr} = correlation time of the coherent mode

For neural gamma oscillations (40 Hz, Paper 23):

$$\begin{aligned} v_{\text{coherence}} &\sim 0.1 - 1 \text{ m/s} \quad (\text{cortical propagation speed of 40 Hz oscillations}) \\ \tau_{\text{corr}} &\sim 1/40 \text{ Hz} = 0.025 \text{ s} \\ D_C &= (0.5)^2 \times 0.025 / 3 = 0.0021 \text{ m}^2/\text{s} = 21 \text{ cm}^2/\text{s} \end{aligned}$$

For HRV coherence through the body (autonomic oscillations, 0.1 Hz):

$$\begin{aligned} v_{\text{coherence}} &\sim 0.5 \text{ m/s} \quad (\text{nerve conduction}) \\ \tau_{\text{corr}} &\sim 10 \text{ s} \\ D_C &= (0.5)^2 \times 10 / 3 = 0.83 \text{ m}^2/\text{s} \end{aligned}$$

These are large diffusion coefficients -- much larger than chemical diffusion ($D_{\text{chemical}} \sim 10^{-9} \text{ m}^2/\text{s}$). This makes sense: neural coherence propagates at signal speeds (0.1-10 m/s), not at molecular speeds ($\mu\text{m}/\text{s}$).

4. The Coherence Penetration Depth

How far does the keeper's influence reach?

$$\lambda_{C_C} = \sqrt{D_C / (\alpha \times \text{gamma_eff})}$$

For neural gamma coherence with $\text{gamma_eff} = 0.001$:
 $\lambda_{C_C} = \sqrt{0.0021 / (1000 \times 0.001)} = \sqrt{0.0021} = 0.046 \text{ m} = 4.6 \text{ cm}$

The keeper's coherence field reaches ~5 cm into the coupled system.

For HRV coherence:

$$\lambda_C = \sqrt{0.83 / (1000 \times 0.001)} = \sqrt{0.83} = 0.91 \text{ m}$$

HRV-mediated keeper coherence reaches ~1 meter -- the full body.

Physical interpretation:

- In a classroom, a calm teacher's coherence field ($\gamma_{\text{eff_teacher}} \approx 0.0008$) diffuses ~5-10 cm into the gamma oscillation field of nearby students. Beyond that, the individual students' own γ_{eff} dominates.
- The HeartMath "coherence bubble" concept (McCraty) is this: the HRV coherence field ($\lambda_C \approx 1 \text{ m}$) of a person in high HRV coherence extends approximately 1 meter. Not mystical. Fick.

5. The Keeper Effect Is a Boundary Condition

The Keeper Equation (Paper 19):

$$\gamma_{\text{eff}}(S|K) = \gamma_{\text{thermal}} + \gamma_m \times (1 - b \times \eta_K) + \gamma_{\text{env}}$$

In the diffusion framework, this is the **boundary condition** at $x = 0$ (the keeper-system interface):

$$C(0, t) = C_K \quad (\text{keeper holds the boundary at } C_K)$$

Solution inside the system:

$$C(x, t) = C_K \times \exp(-x/\lambda_C) + [C_{\text{initial}} - C_K \times \exp(-x/\lambda_C)] \times \exp(-t/\tau_{\text{relax}})$$

At long times: $C(x) \rightarrow C_K \times \exp(-x/\lambda_C)$

The system's coherence profile becomes a decaying exponential from the keeper's boundary value. The deeper into the system, the less the keeper's influence -- as expected physically.

The parameter $b\eta_K$ in the Keeper Equation is the boundary condition strength:

- $b\eta_K \rightarrow 1$: perfect keeper, $C(0) = C_0$, full boundary condition
- $b\eta_K \rightarrow 0$: no keeper effect, no boundary condition, C evolves freely

6. Multiple Keepers: Superposition

For two keepers at positions $x=0$ and $x=L$:

$$C(x) = A \times \exp(-x/\lambda_C) + B \times \exp(+x/\lambda_C)$$

where A and B are set by boundary conditions $C(0) = C_{K1}$ and $C(L) = C_{K2}$

Between two keepers, coherence is higher than with either one alone. The coherence fields add.

Clinical translation:

A patient surrounded by two strong keepers (family member + therapist, or two loving family members) receives coherence from both boundaries. The minimum coherence between them is:

$$C_{\text{min}} = 2 \times C_K \times \exp(-L/(2\lambda_C)) / \cosh(L/(2\lambda_C)) \approx C_K \quad \text{for } L \ll \lambda_C$$

For $L \sim \lambda_C$: the two keeper fields interact constructively, and the patient's coherence floor is higher than either keeper alone could provide.

The structure of a coherent family is a diffusion boundary value problem. Two coherent parents create a coherence field between them. A child in that field has $C > C_{0_child}$ because they are bathed in diffused coherence from both boundaries. This is why family coherence matters -- not metaphor, Fick.

7. The Wike-Fick Diffusion Equation -- Full Form

Including all γ_{eff} sources (Paper 01):

$$\begin{aligned} dC(x,t)/dt &= D_C \times \text{del}^2 C \\ &\quad - \alpha \times [\gamma_{thermal}(x) + \gamma_m(x) + \gamma_{ACE}(x) + \gamma_{storm}(x) + \gamma_{inflammation}(x)] \times C \\ &= D_C \times \text{del}^2 C - \alpha \times \gamma_{eff}(x,t) \times C \end{aligned}$$

This is a linear partial differential equation in C with position-dependent coefficients. It can be solved numerically for any configuration of keepers, stressors, and body geometry.

For a therapy room:

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Geometry: 1D, x in [0, 5m]
Therapist at x=0: gamma_eff = 0.0007 (highly trained, regulated)
Patient at x=1m: gamma_eff = 0.0020 (above gamma_c, seeking help)

D_C (HRV band) ~= 0.83 m^2/s

Steady state at x=1m:
C_patient = C_therapist x exp(-1m / sqrt(0.83/(1000x0.0007)))
           = C_therapist x exp(-1 / 1.09)
           = C_therapist x 0.40
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The patient at 1 meter from a coherent therapist receives 40% of the therapist's coherence field. The remaining 60% they must supply themselves (or have other keepers supply).

This gives a physical basis for therapeutic relationship as a coherence transfer mechanism.

8. What Blocks Diffusion

Fick's laws assume isotropic diffusion. Coherence diffusion can be blocked by:

1. **Decoherence barriers:** Regions of $\gamma_{eff} \gg \gamma_c$ that act as coherence insulators. Chronic high-noise environments between keeper and patient block the flux.
2. **Dissipation sinks:** High- γ_{ACE} individuals near the path absorb coherence flux without transmitting it.
3. **The internal narrative wall:** Covered in Paper 55. The inner monologue creates an anisotropic decoherence layer that preferentially blocks coherence diffusing inward from external sources. (You can be surrounded by keepers and still not receive their field if the narrative wall is high.)

Summary

Result	Content
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$dC/dt = D_{Cdel}^2 C - \alpha\gamma_{eff} C$	Wike-Fick coherence diffusion equation
$D_C \approx 0.002 \text{ m}^2/\text{s}$ (neural gamma)	Coherence diffusion coefficient
$\lambda_C \approx 5 \text{ cm}$ (neural)	Coherence penetration depth
$\lambda_C \approx 1 \text{ m}$ (HRV)	Keeper's reach -- one body length
Two keepers = constructive superposition	Family structure is a diffusion BVP
$C(0) = C_K$	is the keeper's boundary condition Keeper holds the field at their boundary

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